GRADE 10 / MYP 5 • FORCE & MOTION

Modelling motion with graphs

Lesson 3 • Match car motion to position–time and velocity–time graphs.

LEARNING GOALS

Represent and interpret motion

- Plot x–t and v–t graphs with correct units.
- Explain positive, zero and negative gradients.
- Calculate gradients on straight and curved graphs.
- Connect gradients to velocity and acceleration.

READING A MOTION GRAPH

The graph is NOT the road

HORIZONTAL AXIS

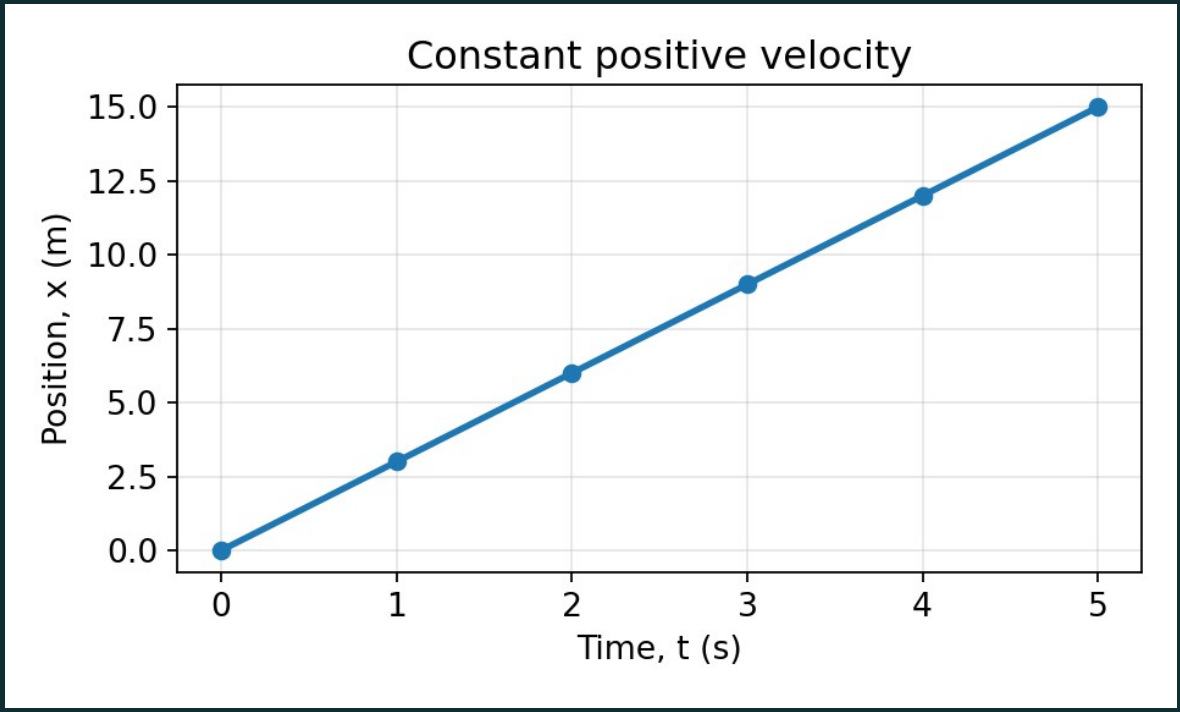
Always shows elapsed time, t (s).

VERTICAL AXIS

Shows position x (m) OR velocity v (m/s).

GRAPH MODELS

Position–time: constant velocity



A constant positive gradient (uphill straight line) means constant velocity in the positive direction.

Slope = Δx / Δt = velocity (m/s).

POSITIVE, ZERO, NEGATIVE

x–t gradients

POSITIVE SLOPE

Car moves in +x.
(Moves with positive velocity.)

Bigger gradient → faster positive velocity.

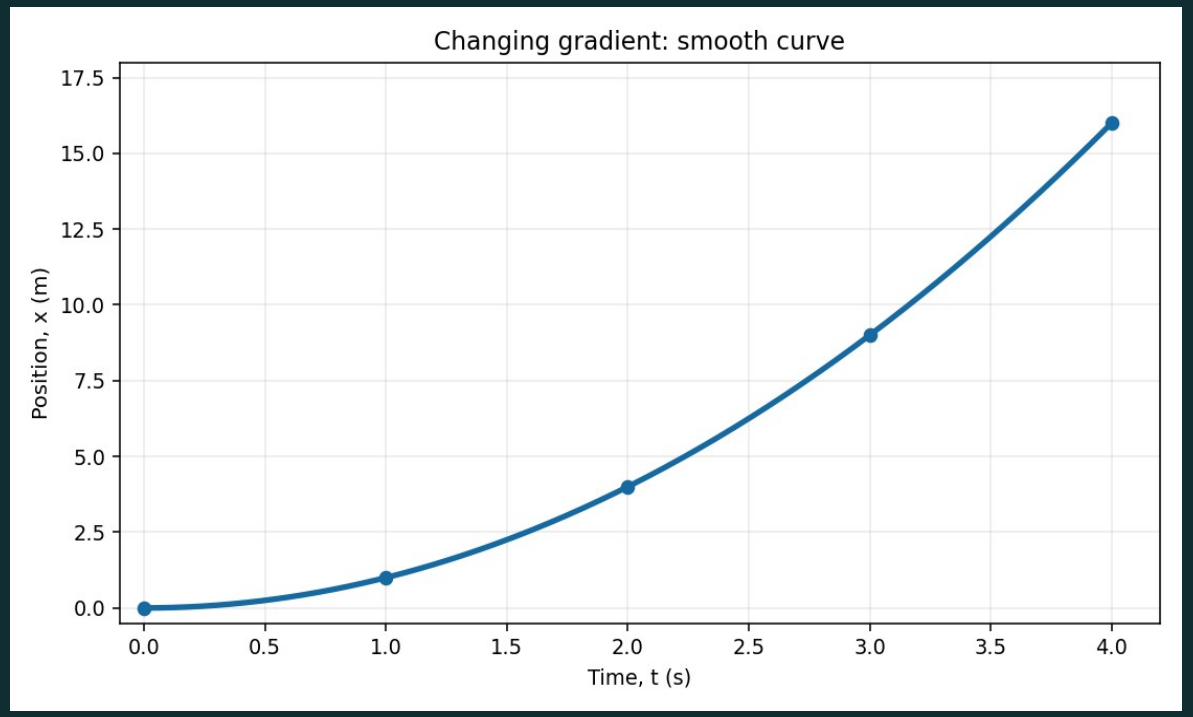
NEGATIVE OR ZERO

Negative: motion in −x.
(Velocity is in the negative direction.)

Zero: position unchanged; car stationary.

GRAPH MODELS

A changing slope



The slope changes with time.

To estimate velocity at one instant, draw a TANGENT at that time and calculate its slope.

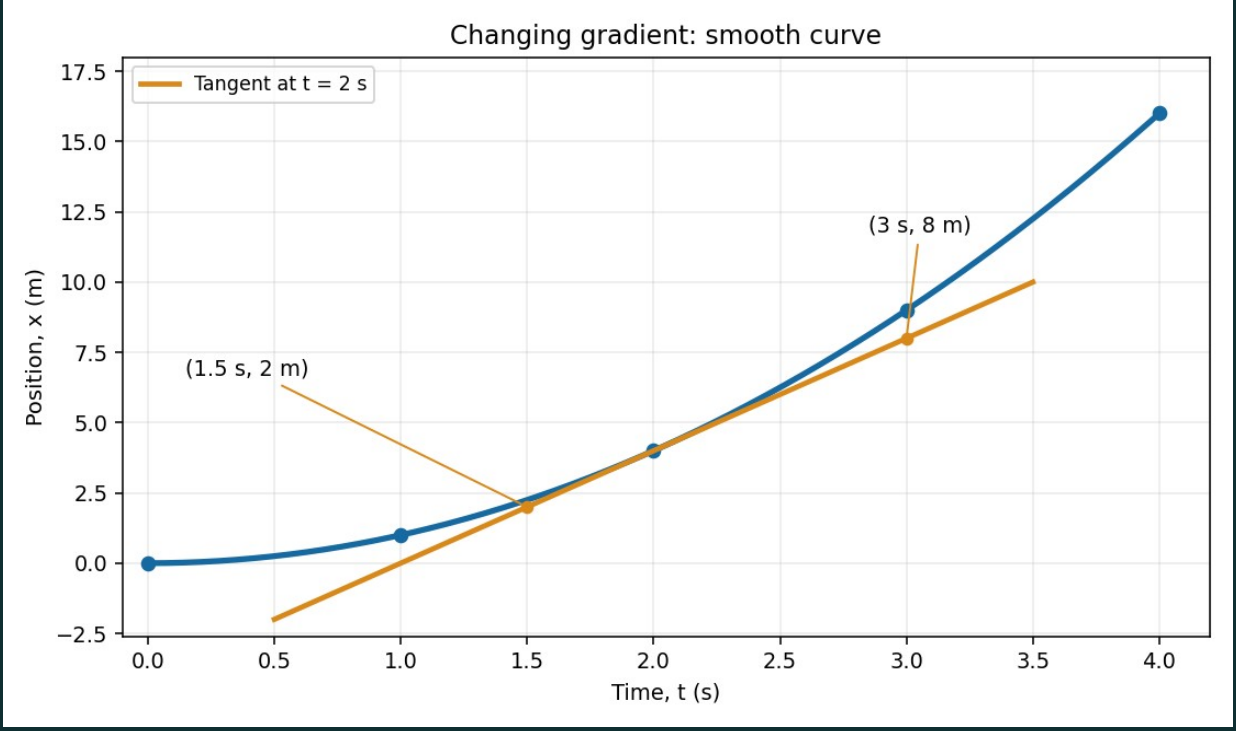
HOW TO FIND A TANGENT GRADIENT

For a curve, not an entire journey

- Mark the instant where you want the velocity.
- Draw a straight line that locally follows the curve.
- Select two well-separated points on this tangent.
- Calculate rise / run, with units m/s.

HOW TO CALCULATE A TANGENT GRADIENT

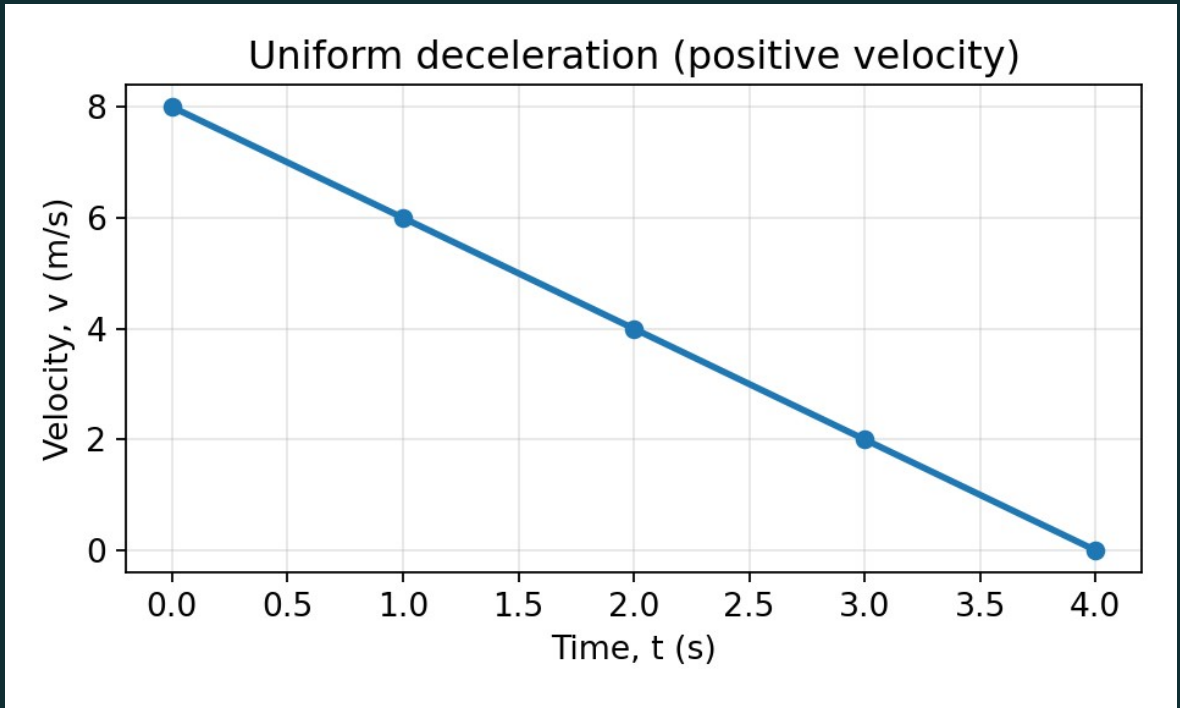
At t = 2 s



Pick two points on the tangent:
(1.5 s, 2 m) and (3.0 s, 8 m)
Rise = 8 − 2 = 6 m
Run = 3.0 − 1.5 = 1.5 s
Velocity = gradient = +4 m/s.

GRAPH MODELS

Velocity–time: slowing down



The car is still moving in +x while velocity drops to zero.

Gradient of v–t = acceleration, in m/s^2.

In this case, acceleration is negative.

In this example, acceleration = ?

A HORIZONTAL GRAPH MEANS...

Different answers on different graphs

HORIZONTAL x–t

No position change.
Velocity = 0.

HORIZONTAL v–t

Constant velocity (can be nonzero!).
Acceleration = 0.

MOTION-SENSOR ACTIVITY

Match the screen trace

- Stand at one known distance from the sensor.
- Match a flat x–t trace; then an upward straight trace.
- Try a downward trace, then a curve that gets steeper.
- Describe the direction, speed and slope in each case.

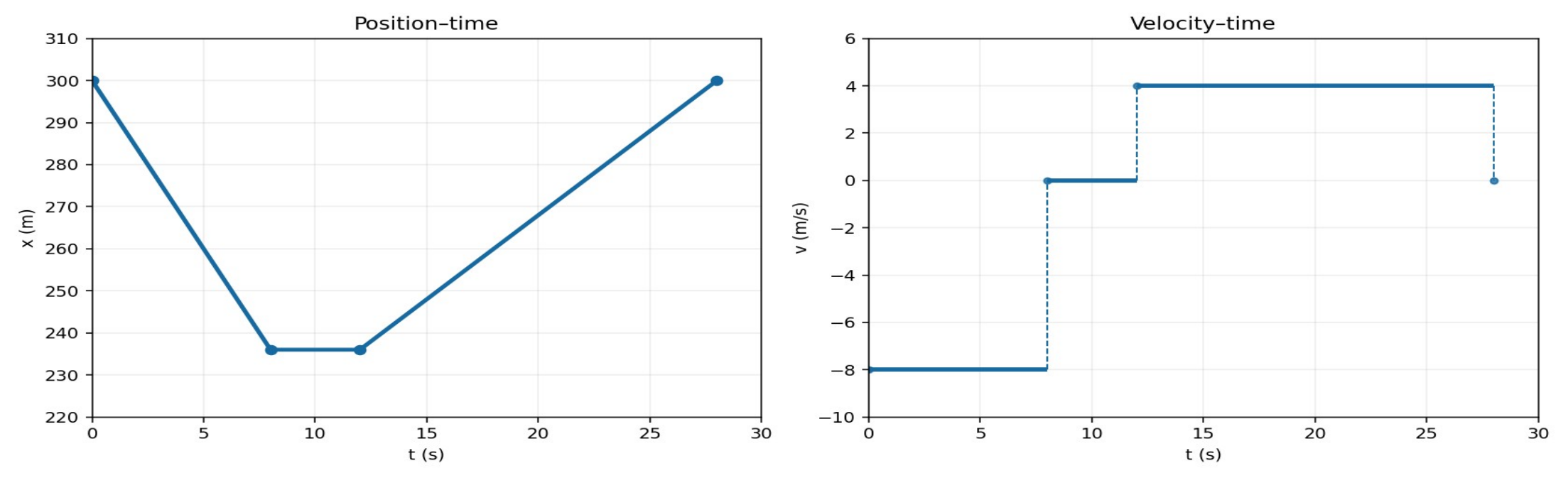
SKETCH FROM A STORY

Discuss before drawing

- A car starts at x = +300 m.
- It travels towards the origin at a constant speed of 8 m/s.
- After exactly 8 s it stops; it remains stationary for a further 4 s.
- It then travels away from the origin at 4 m/s until x = +300 m, where it stops. Sketch the x–t and v–t graphs in your notebook.

SOLUTION — TWO VIEWS OF ONE JOURNEY

Position and velocity against time



How does each graph inform the other? Can you determine additional information from either graph?

GRAPH QUALITY CHECKLIST

Every graph needs…

- Correct quantity on each axis and units.
- Sensible, evenly spaced scale.
- Correctly plotted data; straight best fit or smooth trend.
- Gradient triangle on the chosen line or tangent.

INDEPENDENT PRACTICE

Worksheet + simulation

- Complete graph MCQs and free-response practice.
- Optional homework: Glidepath Science Motion Lab.
- https://glidepathscience.org/simulations/motion-lab/

KEY LEARNING SUMMARY

Two gradients, two meanings

- Gradient of x–t = velocity (m/s).
- Gradient of v–t = acceleration (m/s^2).
- A curve needs a tangent for instantaneous gradient.
- Negative gradient is a direction/sign, not automatically “slower”.
- **Units of a gradient = units of rise ÷ units of run (on any graph).**